

HR Analytics & Workforce Intelligence

Project Overview • Power BI Dashboard • Data Analytics Portfolio Project

1. Project Overview

HR Analytics & Workforce Intelligence is a Power BI project that helps an HR team understand employee data and make data-driven decisions. The dashboard brings employee, recruitment, attendance, leave, performance, training, compensation, and exit information into one interactive report.

2. Project Objective

- Monitor important HR and workforce metrics in one place.
- Understand employee performance, attendance, training, and compensation.
- Analyze recruitment sources and hiring results.
- Identify employee exit and attrition patterns.
- Allow HR users to filter and explore the data interactively.

3. Tools & Technologies

- **Power BI Desktop:** Dashboard creation and data visualization.
- **Power Query:** Data cleaning, transformation, and preparation.
- **DAX:** Measures, KPIs, calculations, and advanced analysis.
- **Excel:** Source dataset and structured tables.
- **SQL:** Used as a supporting data-analysis skill for relational data concepts.
- **Python / Pandas:** Used as a supporting data-analysis skill for data preparation and exploration.

4. Data Model

The project uses a structured model with dimension and fact tables. DimEmployee is connected to employee-related fact tables, while DimDepartment, DimJobRole, DimLocation, and DimDate provide descriptive and time-based information.

- DimEmployee – employee information
- DimDepartment – department information
- DimJobRole – job role information
- DimLocation – location information
- DimDate – date and time analysis
- FactAttendance – attendance records
- FactLeave – leave records
- FactPerformance – performance records
- FactCompensation – salary and bonus information
- FactTraining – training records

- FactRecruitment – recruitment and hiring records
- FactExit – employee exit information

5. Data Cleaning & Preparation

- Corrected text values using Trim and Capitalize Each Word.
- Standardized inconsistent values such as leave and training statuses.
- Corrected spelling inconsistencies in exit-related fields.
- Removed duplicate employee and attendance records where appropriate.
- Converted date and numeric columns to suitable data types.
- Kept meaningful null values, such as missing hiring dates for candidates who were not hired.
- Created relationships between dimension and fact tables.

6. Dashboard Pages

Executive Dashboard – Provides an overall view of employees, attrition, salary, performance, training, recruitment, and attendance.

Recruitment & Hiring – Analyzes applications, hiring, interview scores, recruitment sources, departments, job roles, and hiring trends.

Performance & Employee Development – Shows performance ratings, goal achievement, manager ratings, high performers, promotion eligibility, training hours, scores, and training cost.

Attendance & Leave Analytics – Analyzes attendance rate, absent days, late days, work-from-home days, leave types, leave status, and trends.

Compensation & Salary Analytics – Shows salary, bonus, salary hike, department/job-role comparisons, trends, and projected salary using a What-if parameter.

Employee Overview & Workforce Demographics – Shows workforce size, active employees, gender, job level, department, location, age distribution, and workforce trends.

HR Insights & Advanced Analytics – Analyzes exits, exit reasons, exit types, attrition trends, and factors related to employee exits.

Employee Details – Provides employee-level information through drill-through, including salary, performance, training, leave, and basic employee details.

7. Power BI Features Used

- DAX measures for KPIs and business calculations.
- Slicers for interactive filtering.
- What-if parameter for salary projection.
- Field parameter for switching compensation metrics.
- Drill-through for employee-level analysis.
- Report-page tooltip for additional employee information.
- Decomposition Tree for exploring employee exits.
- Key Influencers for analyzing factors associated with employee exits.
- Row-Level Security (RLS) using the HR Department role.

8. Important KPIs

Area	Examples of KPIs
Workforce	Total Employees, Active Employees
Recruitment	Total Applications, Hired Candidates, Hiring Rate
Performance	Average Performance Rating, High Performers
Attendance	Attendance Rate, Absent Days, Late Days
Training	Training Hours, Training Score, Training Cost
Compensation	Average Salary, Total Salary, Total Bonus, Salary Hike
Attrition	Total Exits, Attrition Rate, Average Notice Period

9. Sample Dashboard Results

- The model contains 500 employees.
- The dashboard shows an overall attrition rate of 20.0% for the available project data.
- The recruitment analysis contains approximately 5,000 applications.
- The model contains attendance, leave, performance, compensation, training, recruitment, and exit records for analysis.
- The Compensation page includes an interactive salary projection using a user-selected salary increase percentage.

10. Security & Advanced Analysis

Row-Level Security was implemented with an HR Department role. The role filters DimDepartment to the HR department so that the report can demonstrate department-level access control.

11. Project Outcome

This project demonstrates an end-to-end Power BI workflow: preparing data, building a relational data model, creating DAX measures, designing interactive dashboards, and using advanced Power BI features for HR analysis. It can help HR teams monitor workforce metrics and explore areas such as recruitment, performance, attendance, compensation, training, and attrition.

12. Project Learning

- Data cleaning and transformation using Power Query.
- Data modeling using relationships between dimension and fact tables.
- Writing DAX measures for business KPIs.
- Designing interactive and user-friendly dashboards.
- Using advanced Power BI features for deeper analysis.
- Applying Row-Level Security for controlled access.

Note: This documentation is a simple project overview. The PBIX file contains the complete Power BI model, measures, relationships, report pages, and interactive features.